MACHINE
LEARNING

ASSIGNMENT 2

Submitted By:

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GitHub Repository: https://github.com/Aiman-rafiq/Zee-Outsourcing-Internship-ML/tree/main/Week-02-Data-Preparation

1. Introduction

The objective of this assignment is to prepare a raw dataset for machine learning through exploratory data analysis, data cleaning, feature engineering, and preprocessing.

The Student Performance Factors dataset was selected to examine how academic, personal, and environmental factors relate to students' examination results. The final goal is to produce clean, encoded, and scaled training and testing datasets suitable for machine learning.

2. Dataset Description

Attribute	Details
Dataset Name	Student Performance Factors
Source	Kaggle — lainguyn123
Total Records	6,607
Original Columns	20
Numerical Columns	7
Categorical Columns	13
Target Variable	Exam_Score
Problem Type	Regression

Dataset Source: https://www.kaggle.com/datasets/lainguyn123/student-performance-factors

The dataset contains information about study hours, attendance, parental involvement, motivation, previous scores, sleep hours, and other factors potentially associated with academic performance.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset structure, examine numerical distributions, identify missing values, and explore relationships between features and examination scores.

3.1 Distribution of Exam Scores

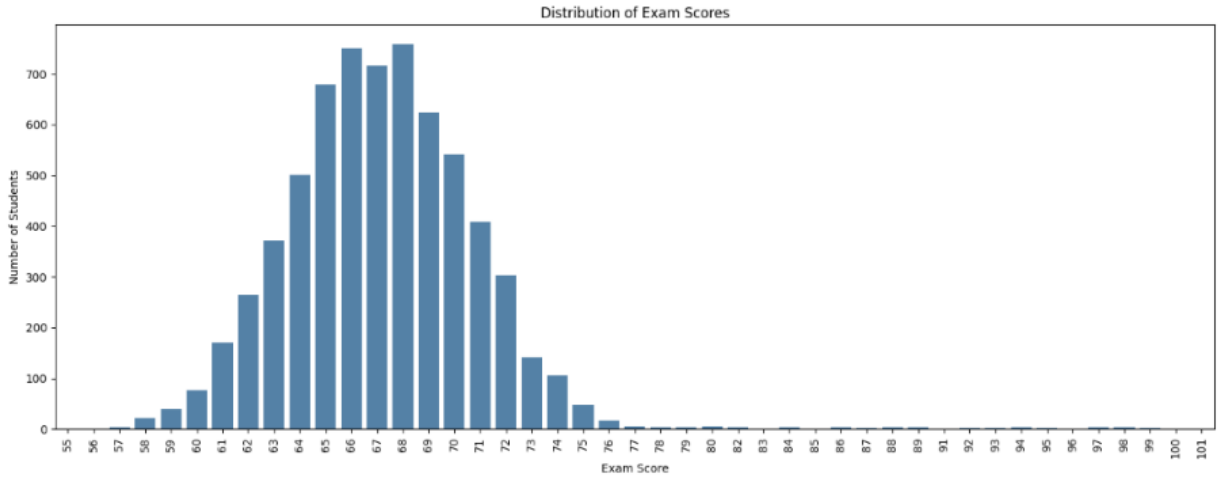


Figure 1: Distribution of students' examination scores.

Observation: Most students scored between approximately 64 and 71 marks, with the highest concentration around 66–68. The distribution is roughly bell-shaped, with relatively few students obtaining very high scores. An unusual maximum score of 101 was also observed and flagged for further investigation.

3.2 Correlation Analysis

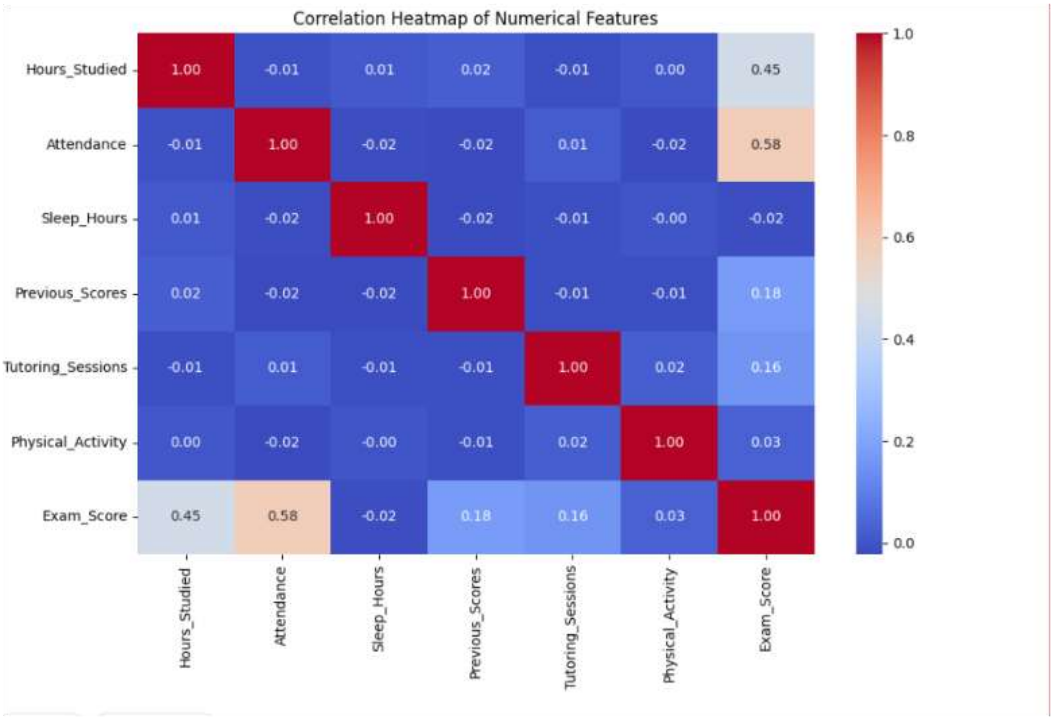


Figure 2: Correlations between numerical features and Exam_Score.

Observation: Attendance shows the strongest positive correlation with examination scores (0.58), followed by Hours_Studied (0.45). Previous_Scores (0.18) and Tutoring_Sessions (0.16) have weaker positive relationships. Sleep_Hours and Physical_Activity show almost no linear correlation with the target.

These findings suggest that attendance and study hours may be useful predictors.

4. Data Cleaning

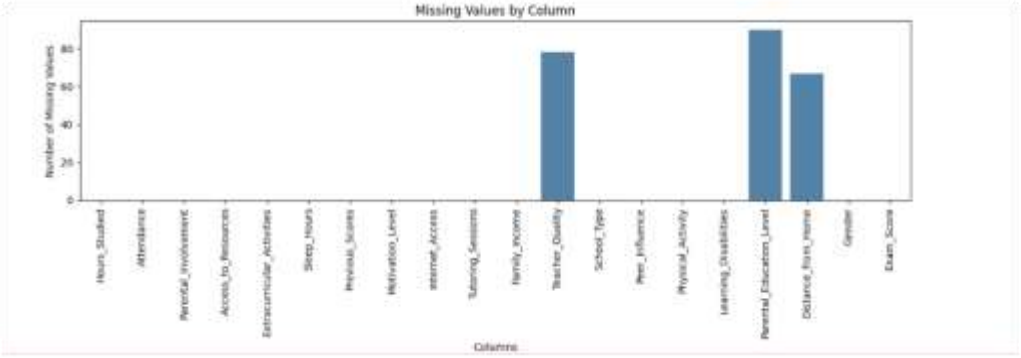
Data cleaning was performed to identify duplicate records, assess missing values, and review unnecessary columns.

No duplicate records were removed. Three categorical columns contained missing values. Rather than removing student records, the decision was made to replace missing values with the most frequent category, using values learned only from the training data.

Cleaning Decisions

Column	Problem	Action	Reason
Teacher_Quality	78 missing values	Mode imputation	Preserve useful categorical information
Parental_Education_Level	90 missing values	Mode imputation	Avoid unnecessary record deletion
Distance_from_Home	67 missing values	Mode imputation	Retain a potentially useful feature
Duplicate Records	None detected	No removal needed	All original records retained
ID/Name Columns	Not present	No columns dropped	No obvious identifiers
Exam_Score	Maximum value 101	Retained and flagged	Requires verification of scoring rules

Missing values were intentionally filled during preprocessing after the train/test split to avoid data leakage.



5. Feature Engineering

Two new numerical features were created using existing variables.

Feature 1: Study_Attendance_Interaction

Formula: Hours_Studied × Attendance / 100

This feature combines students' study effort with their classroom attendance. It may help the model identify patterns involving both independent factors.

Feature 2: Study_Sleep_Ratio

Formula: Hours_Studied / Sleep_Hours

This feature represents the relationship between study time and sleep duration. It may help the model recognize different study and rest patterns.

After feature engineering, the dataset contained 22 columns, including the target variable. Neither engineered feature used the target value.

6. Train-Test Split

The dataset was divided into 80% training data and 20% testing data using random_state=42 to ensure reproducibility.

Detail	Training	Testing
Records	5,285	1,322
Features before encoding	21	21
Mean Exam Score	67.22	67.32
Median Exam Score	67	67

Stratification was not used because Exam_Score is a numerical regression target rather than a categorical class label. Class proportions are therefore not applicable.

The similar training and testing target averages suggest that the split produced reasonably comparable target distributions.

7. Data Preprocessing

Preprocessing was performed using Scikit-learn Pipelines and ColumnTransformer to prepare the features for machine learning.

Numerical Features:

- SimpleImputer with the median strategy was used to handle missing numerical values.
- StandardScaler was used to standardize numerical features, helping ensure that differences in feature scales do not disproportionately influence scale-sensitive models.

Categorical Features:

- SimpleImputer with the most frequent strategy was used to replace missing categories.
- OneHotEncoder converted categorical values into numerical indicator columns. The handle_unknown="ignore" setting allowed previously unseen categories to be handled during transformation.

Data Leakage Prevention

The preprocessing pipeline was fitted exclusively on training data using fit_transform(X_train).

Testing data was processed using transform(X_test) without fitting again.

As a result, the test data did not influence the learned medians, most frequent categories, scaling parameters, or encoding categories.

8. Final Results

After preprocessing, all feature columns were numerical and contained no missing values.

Final Output	**Shape**
X_train	(5285, 42)
X_test	(1322, 42)
y_train	(5285,)
y_test	(1322,)

Final Validation:

- Missing values in X_train: 0
- Missing values in X_test: 0
- Missing values in y_train: 0
- Missing values in y_test: 0
- All input features are numerical.
- Preprocessing was fitted only on training data.

The increase from 21 input features to 42 processed features resulted from one-hot encoding of categorical variables.

9. Challenges and Learning Outcomes

One challenge was understanding how numerical and categorical features require different preprocessing techniques. Another was ensuring that missing-value imputation and scaling did not introduce data leakage.

I also observed an unusual maximum exam score of 101, highlighting the importance of reviewing unexpected values rather than automatically deleting them.

Through this assignment, I learned how to explore a dataset, handle missing information, engineer new features, separate training and testing data, and create reusable preprocessing pipelines.

10. Conclusion

The Student Performance Factors dataset was successfully prepared for machine learning. The final output consists of X_train, X_test, y_train, and y_test, with 42 clean numerical input features and zero missing values. These datasets are ready for use in a subsequent machine learning modeling stage.